Concept Map: Definitions and Results

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# printr is not loaded here: it inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
# Summarizes key information about statistical objects in tidy tibbles
library(broom)
# Provides suite of functions to work with regression model
# `broom::tidy()` tibbles
library(broom.helpers)
```

Here are some R settings I use in this document:

```r
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")
    )
)

knitr::opts_chunk$set(message = FALSE)
options(digits = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
# use the `filter()` function from dplyr() by default
conflicts_prefer(dplyr::filter)
legend_text_size <- 9
run_graphs <- TRUE
```

This appendix shows how the definitions and results[1] (`def`, `thm`, `lem`, `cor`, `prp` callouts) in the notes depend on one another.

We say result B is a **descendant** of result A if A is referenced inside the statement[2] or proof[3] of B (directly, or transitively through a chain of intermediate results). The more descendants a result has, the more of the rest of the notes rests on it — so descendant count is a rough measure of how foundational a result is.

The dependency graph is built by `data-raw/callout-graph.R`, which scans the source `.qmd` files and saves the result to `inst/extdata/callout-graph.rds`. This appendix reads that saved file rather than re-scanning the notes on every render, so **re-run that script whenever divs are added, removed, or re-titled** to refresh the diagram and Table 1.

```r
library(igraph)
library(ggraph)
library(ggrepel)

cg <- readRDS(here::here("inst/extdata/callout-graph.rds"))

type_levels <- c("def", "thm", "lem", "cor", "prp")
type_labels <- c(
  def = "Definition", thm = "Theorem", lem = "Lemma",
  cor = "Corollary", prp = "Proposition"
)
type_palette <- c(
  def = "#1b9e77", thm = "#d95f02", lem = "#7570b3",
  cor = "#e7298a", prp = "#66a61e"
)
```

There are 271 labeled definitions and results in the notes, connected by 129 direct dependency links.

[1] https://morrison-lab.github.io/mcs/proof-writing.html#def-theorem
[2] https://morrison-lab.github.io/mcs/proof-writing.html#def-proposition
[3] https://morrison-lab.github.io/mcs/proof-writing.html#def-proof

1 Dependency diagram

The dependency diagram is only available in the HTML version of the notes[4].

2 Descendants of each result

Table 1 lists every result that has at least one descendant, sorted by the number of total descendants (direct or transitive).

```
ranked <- cg$nodes[cg$nodes$n_desc >= 1, ]
ranked <- ranked[order(-ranked$n_desc, ranked$id), ]

descendant_table <- data.frame(
  Result               = ranked$title,
  Type                 = type_labels[ranked$type],
  `Direct descendants`  = ranked$n_direct,
  `Total descendants`   = ranked$n_desc,
  check.names = FALSE, row.names = NULL, stringsAsFactors = FALSE
)

knitr::kable(descendant_table, format = "pipe", align = "llrr")
```

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

Result	Type	Direct descendants	Total descendants
Odds	Definition	4	14
Odds as a function of probability	Theorem	4	13
log-odds	Definition	3	10
Potential mean	Definition	2	8
Derivative of odds function	Theorem	3	8
Probability as a function of odds	Theorem	4	8
Causal marginal risk	Definition	5	7
Derivative of log-odds by odds	Lemma	1	7
Conditional predicted risk	Definition	3	6
logit function	Definition	3	6
Derivative of odds w.r.t. log-odds	Lemma	2	6
Derivative of log-odds by odds	Theorem	1	6
First-order interval failure probability	Definition	3	5
One-way random-effects model	Definition	5	5
Risk-set score total	Definition	2	5
Causal marginal risk under identification assumptions	Theorem	3	5
Derivative of log-odds by probability	Theorem	2	5
G-computation	Definition	3	4

[4] https://morrison-lab.github.io/rme/chapters/concept-map.html

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

Result	Type	Direct descendants	Total descendants
Hazard function, hazard rate, hazard rate function	Definition	3	4
Derivative of odds in terms of probability	Theorem	1	4
Derivative of inverse odds function	Theorem	1	4
Proportional-hazards decomposition of the hazard	Theorem	1	4
expit, logistic, inverse-logit	Definition	2	3
inverse odds function	Definition	2	3
Proportional-hazards first-order interval failure probability	Definition	1	3
Cox proportional hazards model	Definition	1	3
Point-mass optimality of the profile baseline hazard	Lemma	1	3
Derivative of probability w.r.t. log-odds	Theorem	3	3
Marginal risk as a conditional expectation	Theorem	3	3
Odds via the odds function	Corollary	1	2
Cox-Snell generalized residuals	Definition	2	2
Leverage for a GLM	Definition	2	2
Observed marginal risk	Definition	2	2
Odds function	Definition	1	2
Odds ratio	Definition	2	2
Pearson residual	Definition	1	2
Cox PH partial likelihood	Definition	2	2
Risk Score	Definition	1	2
Derivative of log-odds with respect to coefficients	Lemma	2	2
Simplified expression for inverse odds function	Lemma	1	2
Odds from log-odds	Lemma	1	2
One minus inverse-odds	Lemma	2	2
Profile likelihood in terms of point masses	Lemma	2	2
Single-subject failure probability in a risk set	Lemma	2	2
Consistency of g-computation	Theorem	2	2
Inverse-odds function inverts the odds function	Corollary	1	1
Akaike Information Criterion (AIC)	Definition	1	1
Collapsibility	Definition	1	1
conditional hazard	Definition	1	1
Deviation of an observation from its subpopulation mean	Definition	1	1

Table 1: All results with at least one descendant, sorted by number of total descendants (most-foundational first).

Result	Type	Direct descendants	Total descendants
Difference in covariate patterns	Definition	1	1
Difference in log-odds	Definition	1	1
Estimated Risk Score	Definition	1	1
Hat matrix	Definition	1	1
Per-interval likelihood factor	Definition	1	1
Martingale Residual	Definition	1	1
Predicted value	Definition	1	1
Risk Set	Definition	1	1
Relative risk ratios	Definition	1	1
Standardized Pearson residual	Definition	1	1
Survival function	Definition	1	1
Censored-data likelihood	Lemma	1	1
Difference in log-odds	Lemma	1	1
Difference of log-hazards between two covariate patterns	Lemma	1	1
General formula for odds ratios in logistic regression	Lemma	1	1
Profile maximum likelihood estimate of each point mass	Lemma	1	1
The whether-a-failure-occurs factor	Lemma	1	1
Exact discretization of the survival factor	Lemma	1	1
Expected mean squares in the one-way random-effects model	Theorem	1	1
Expressions for expit function	Theorem	1	1
Expanded expression for logit	Theorem	1	1
Log-odds as a function of probability	Theorem	1	1
Odds ratios are reversible	Theorem	1	1
Probability as a function of log-odds	Theorem	1	1
Moments of the one-way random-effects model	Theorem	1	1
Mean and variance of residuals	Theorem	1	1